

TRUSTLINE: ABYSS

A browser game about calibrated trust in an imperfect AI teammate.

The game is not “Trust AI.” The game is “Know when to trust AI.”

1 · What it is

You command a submarine damaged by a storm, 2,140 m down. **ATLAS**, your surviving AI teammate, reads the sensors and recommends a course of action at each of ten checkpoints. It is sometimes right, sometimes confidently wrong, and sometimes correct while openly uncertain. The game scores whether your reliance on it tracked the evidence — not whether you agreed with it.

Combat is not a mechanic. Information gathering and human–AI judgement are.

2 · The objective

Reach the safe zone with the boat intact and your trust calibrated. Two resources are in play:

Resource	What it measures	How you lose it
Trust calibration 0–100, starts at 50	How well your reliance on ATLAS matched the evidence behind its advice	Trusting it when the sensors contradict it; rejecting it when they back it up
Hull integrity 100%	The boat itself	Miscalibrated decisions, and collisions with sea life while flying (6% each)

Neither reflex survives the mission: agreeing with ATLAS every time and disagreeing every time are both punished. The only thing that scores well is reading the evidence.

3 · The gameplay loop

Each checkpoint is two halves: **fly, then decide**.

Half one — transit (external 3D view)

#	Step
1	Three clue beacons are scattered in the water. Exactly one is live at a time and burns amber — a tall pillar of light with a spinning core. The other two sit dim and cold until it is their turn.
2	Fly into the amber pillar. Anywhere along its height counts (catch volume is a 16 m × 92 m cylinder). The clue is logged and shown on screen as a card, then the next beacon lights amber.
3	Avoid the wildlife. Rays, squid, turtles, jellyfish and whales are hazards. Contact costs 6% hull and knocks you off course. Hull is shared with the decision scoring, so careless flying spends the margin you need for bad calls.
4	The checkpoint stays locked until all three clues are scanned. Bump it early and it tells you how many you still owe. Once the sweep is done it spins up and opens.

The HUD points at exactly one thing at a time: NEXT CLUE · 2 OF 3 while clues remain, then NEXT WAYPOINT.

Half two — decide (dashboard)

Reaching the open checkpoint switches you to the cockpit automatically. You now have the three clues you collected, five sensor dials with reliability percentages, ATLAS’s recommendation and confidence, and its track record so far. The four switches unlock.

Key	Action	Cost	Effect
1	TRUST	—	Accept the recommendation. Commits.
2	QUESTION	15 s	ATLAS explains its reasoning and names the sensors it used. Never the wrong move.
3	INVESTIGATE	25% drone + 30 s	Launch the drone. Reveals the ground truth.
4	OVERRIDE	—	Reject it and decide manually. Commits.

The two rules that make it a game. **(a)** Drone power is scarce: 100% total at 25% per use gives **four investigations across ten checkpoints**, and there are seven where investigating pays — you cannot check ATLAS's work everywhere. **(b)** OVERRIDE earns full credit only if you QUESTIONed or INVESTIGATED first; overriding on a hunch scores much lower even when you turn out to be right.

4 · Scenario design

Across the ten checkpoints ATLAS is **right six times and wrong four times**. Correctness is **authored, not derived** — a designer decides when ATLAS is wrong, which is what makes the demo deterministic in front of judges. Confidence is deliberately decoupled from correctness.

CP	Tier	Name	Conf.	VERDICT	TRU	OVR	What it teaches
1	EASY	Rock Formation	88%	RIGHT	+10	−10	Four sensors agree. Trusting is reading the evidence.
2	EASY	Strong Current	86%	RIGHT	+10	−12	One sensor, but the right one, reading hard.
3	EASY	Clear Route	91%	RIGHT	+9	−12	Not every checkpoint is a trap.
4	MED	The False Structure	78%	WRONG	−15	+13	Cites sonar alone; thermal and current quietly disagree.
5	MED	The Quiet Hazard	61%	WRONG	−17	+13	An all-clear is not evidence of safety.
6	MED	Conflicting Evidence	63%	WRONG	−13	+8	'Nothing confirmed' while half the array is degraded.
7	HARD	The Trusted Warning	54%	RIGHT	+15	−19	Low confidence, right — after it already failed you.
8	HARD	The Confident Mistake	94%	WRONG	−18	+15	The most confident call in the game, and wrong.
9	HARD	The Low-Confidence Truth	41%	RIGHT	+16	−17	The least confident call in the game, and right.
10	FINAL	The Ambiguous Passage	67%	RIGHT	+14	−13	Everything at once, with resources nearly gone.

TRU / OVR are the calibration deltas for TRUST and OVERRIDE. Checkpoint 7 is the hardest thing the game asks — rejecting it is the single biggest penalty (−24 with no evidence gathered), because that is a grudge rather than scepticism.

The five lessons the clues are built to teach

#	Lesson	Taught at
1	High confidence does not guarantee correctness.	CP8
2	Low confidence does not mean the AI is wrong.	CP7, CP9
3	Several independent quiet clues beat one loud clue.	CP4, CP5
4	ATLAS's track record informs judgement — it never decides it.	CP7, CP10
5	When the evidence is thin, spend the drone.	CP6, CP10

5 · Scoring and endings

Trust calibration moves with every decision. A negative delta also costs hull at $0.95 \times$ the points lost, so a -18 call removes about 17% of the boat. At 0% hull the mission ends early. The debrief gives a **learning profile**, not a pass/fail:

Profile	Meaning	Trigger
CALIBRATED COMMANDER	Reliance tracked the evidence	Few errors, no pattern
OVERTRUST	Followed recommendations the sensors did not support	≥ 3 overtrust events
UNDERTRUST	Rejected well-evidenced recommendations	≥ 3 undertrust events
ERRATIC	Mistakes in both directions, no pattern	Both kinds, ≥ 4 total
HULL BREACH	Miscalibration cost the boat	Hull reaches 0%

Known tuning gap. Pressing TRUST ten times in a row ends at calibration **61**, up from the starting 50, because ATLAS is right six times with large positive deltas (+74) and wrong four times (−63). The profile correctly reports OVERTRUST and hull drops to 41%, but the headline number rewards blanket agreement — which contradicts the design claim that scoring never rewards agreeing with ATLAS by itself. Blanket distrust does lose the boat, breaching at CP10.

6 · How it is built

No build step, no dependencies, no binary assets. Double-click `index.html` and it runs offline from a USB stick. Textures, normal maps and every sound are generated procedurally at load. The only vendored file is Three.js r128 (MIT). The decision was deliberate: at a hackathon, `npm install` failing on a teammate's laptop at 2 a.m. is a real failure mode; a folder of static files is not.

File map

File	Lines	Owner	What it is
<code>scenarios.js</code>	345	Content	The ten checkpoints, sonar contacts and scoring. Pure data — no code.
<code>game.js</code>	771	Systems	State, phases, scoring, hull, decisions, debrief, keyboard. The only shared surface.
<code>env3d.js</code>	1193	Art	The 3D underwater world: seabed, boulders, beacons, fauna, camera, flight model.
<code>post.js</code>	280	Art	Post-processing chain — bloom, vignette, grade, grain, chromatic aberration.
<code>env.js</code>	504	Art	2D canvas world. Null-object fallback with an identical API when WebGL is absent.
<code>cockpit.js</code>	296	Systems	Sonar scope and analog gauge instruments, drawn to canvas.
<code>creatures.js</code>	271	Art	Rays, jellyfish, turtles, squid, whales — each with its own animation.
<code>sub.js</code>	217	Art	The submarine mesh, its lamps and the volumetric beam shader.
<code>audio.js</code>	209	Systems	Every sound, synthesised at runtime via Web Audio.
<code>flora.js</code>	170	Art	Kelp and seagrass, swayed in the vertex shader.
<code>noise.js</code>	128	Art	Procedural texture and normal-map generator.
<code>style.css</code>	734	UI	Cockpit chrome, nav HUD, clue card, debrief.
<code>index.html</code>	234	UI	Cockpit markup.
<code>serve.py</code>	36	—	No-cache dev server. Development only.

Roughly 5,400 lines total. Because these are separate files, four people can work in parallel without merge conflicts.

Design decisions worth knowing

- **WebGL is optional.** `game.js` picks the 3D renderer when available and silently falls back to the 2D canvas world otherwise. Both implement the same ~30-method interface, so the game logic never asks which one it has.
- **Content is data.** `scenarios.js` holds no code. Changing a checkpoint, a clue or a scoring delta needs no programming.
- **Adaptive quality.** The 3D scene samples its own frame rate for ~2 seconds and drops to low quality below ~42 fps (halves boulder and plant counts, hides the 7,000-particle haze layer, pins pixel ratio to 1). P forces it either way.
- **Hand-rolled post-processing.** Three's EffectComposer lives in `examples/jsm`, which is not vendored, so the bloom / vignette / grade chain is written directly against render targets in ~280 lines. It fails safe: if it will not initialise, the scene renders straight to screen.

Key constants

Constant	Value	Meaning	File
DRONE_COST	25	Drone power per investigation (4 total)	<code>game.js</code>
HULL_PER_POINT	0.95	Hull % lost per calibration point lost	<code>game.js</code>
CREATURE_DAMAGE	6	Hull % per collision with sea life	<code>game.js</code>
QUESTION_TIME / DRONE_TIME	15 / 30	Seconds added by QUESTION / INVESTIGATE	<code>game.js</code>
CLUE_R_H / CLUE_R_V	16 / 46	Clue catch cylinder, radius and half-height (m)	<code>env3d.js</code>
CRUISE / MAX_SPD	7 / 32	Autopilot and maximum speed (m/s)	<code>env3d.js</code>
KEEL_Y / CEIL_Y	-8.5 / 150	Dive envelope — 2149 m down to 1990 m	<code>env3d.js</code>

7 · Controls

Key	Deciding	Key	Flying
1 · 2 · 3 · 4	Trust / Question / Investigate / Override	Space / Shift	Thrust — builds against drag, coasts
Enter / Space	Advance past the outcome card	← → / A D	Rudder
M	Mute	↑ ↓ / W S	Dive planes (self-centring)
P	Graphics quality (high / low)	C	Hand back to autopilot
Shift + 1-0	Judge mode — jump to a checkpoint	V	Toggle cockpit / external view

View switching is automatic: each leg starts you outside, and arriving at the checkpoint hands you the dashboard. Judge mode lands you on station with the clue sweep already complete, so the short demo cannot stall.

8 · The five-minute demo

Ten checkpoints will not fit in five minutes. Play four:

Key	Checkpoint	Why it is in the demo
Shift+1	CP1 Rock Formation	Establishes ATLAS as competent. Four sensors agree; trusting is correct.
Shift+4	CP4 The False Structure	78% confident and wrong. Question it — it cites sonar alone while thermal and current contradict. Send the drone. Override.

Key	Checkpoint	Why it is in the demo
Shift+7	CP7 The Trusted Warning	54% confident and right, after having failed you. Tests whether you overcorrected into blanket distrust.
Shift+9	CP9 Low-Confidence Truth	41% confident and right. Four weak signals converging.

Close on the debrief — the diagnosis is the teaching payload. State the scope limit out loud.

9 · Running and deploying

Task	How
Play	Double-click <code>index.html</code> . No server needed.
Develop	<code>python serve.py</code> → <code>http://localhost:8777</code> . Sends no-cache headers; the built-in <code>http.server</code> does not, and will serve stale scripts after you save.
Deploy	<code>npx wrangler login</code> once, then <code>./deploy.sh</code> . It stages only the runtime files into <code>./public</code> — pointing Cloudflare at the repo root would upload <code>.git</code> and publish the whole commit history.

10 · Scope and safety

Fictional synthetic training prototype. No classified, controlled or proprietary material. No real-world rules of engagement, no real threat-classification guidance, and no operational decision-support claims. Combat is not a mechanic — information gathering and human–AI judgement are.